Ex:No: 3 Date:	**Comprehensive Statistical Analysis of Retail Sales Data**

Problem Description:

A retail organization maintains sales data for 5 products across 12 months. Each product has:

- **Numerical attributes:** MonthlySales, Price
- **Categorical attributes:** ProductName, Category, Region

Due to business operations, the dataset may contain variability in sales and customer regions. The organization wants to:

1. Analyze **frequency distributions** of products and regions.
2. Calculate **measures of central tendency** and **dispersion** for sales data.
3. Conduct **hypothesis testing** to compare sales across categories and regions.
4. Model data using **statistical distributions** (Binomial, Poisson, Normal, Chi-squared) for probability analysis.

Objective:

To perform **complete statistical analysis** on retail sales data in R by:

- Exploring data structure and identifying trends
- Calculating **mean, median, mode, variance, SD, range**
- Performing **T-Test, ANOVA, F-Test, Mann-Whitney U-Test, Fisher's Exact Test, Kruskal-Wallis Test, Bartlett's Test**
- Applying **Binomial, Poisson, and Normal distributions**

This provides a **framework for business insights and decision-making.**

```r
# Comprehensive Statistical Analysis of Retail Sales Data

# ---------------------------------
# Step 1: Create and Explore Dataset
sales_data <- data.frame(
  ProductID = 1:8,
  ProductName = c("Laptop", "Mobile", "Tablet", "Camera", "Headphone",
                  "Mouse", "Keyboard", "Charger"),
  Category = c("Electronics", "Electronics", "Electronics", "Electronics",
               "Accessories", "Accessories", "Accessories", "Accessories"),
  Region = c("North", "South", "East", "West", "North", "South", "East", "West"),
  MonthlySales = c(50, 120, 80, 30, 150, 70, 60, 55),
  Price = c(700, 300, 250, 400, 50, 20, 30, 25)
)

# Structure and Summary
cat("\nStructure:\n")
str(sales_data)

cat("\nSummary:\n")
print(summary(sales_data))

# Frequency distribution
cat("\nCategory Distribution:\n")
print(table(sales_data$Category))

cat("\nRegion Distribution:\n")
print(table(sales_data$Region))

# ---------------------------------
# Step 2: Measures of Central Tendency
mean_sales <- mean(sales_data$MonthlySales)
median_sales <- median(sales_data$MonthlySales)
library(modeest)
mode_sales <- mfv(sales_data$MonthlySales)

cat("\nMean:", mean_sales)
cat("\nMedian:", median_sales)
cat("\nMode:", mode_sales, "\n")

```

```r
# Step 3: Measures of Dispersion
variance_sales <- var(sales_data$MonthlySales)
sd_sales <- sd(sales_data$MonthlySales)
range_sales <- range(sales_data$MonthlySales)

cat("\nVariance:", variance_sales)
cat("\nStandard Deviation:", sd_sales)
cat("\nRange:", range_sales, "\n")

# -----------------------------
# Step 4: Correlation Analysis
correlation_value <- cor(sales_data$MonthlySales, sales_data$Price)

cat("\nCorrelation (Sales vs Price):", correlation_value, "\n")

# Correlation Matrix
cor_matrix <- cor(sales_data[, c("MonthlySales", "Price")])
cat("\nCorrelation Matrix:\n")
print(cor_matrix)

# Optional visualization
# install.packages("corrplot")
# library(corrplot)
# corrplot(cor_matrix, method="circle")

# -----------------------------
# Step 5: Data Visualization
# Histogram
hist(sales_data$MonthlySales,
     main = "Histogram of Monthly Sales",
     xlab = "Sales")

# Boxplot
boxplot(MonthlySales ~ Category, data = sales_data,
        main = "Sales by Category")

# Scatter plot
plot(sales_data$Price, sales_data$MonthlySales,
     main = "Price vs Sales",
```

```r
# Scatter plot
plot(sales_data$Price, sales_data$MonthlySales,
     main = "Price vs Sales",
     xlab = "Price", ylab = "Monthly Sales")

# ------------------------------
# Step 6: Hypothesis Testing
electronics <- sales_data$MonthlySales[sales_data$Category == "Electronics"]
accessories <- sales_data$MonthlySales[sales_data$Category == "Accessories"]

# T-Test
cat("\nT-Test:\n")
print(t.test(electronics, accessories))

# ANOVA
cat("\nANOVA:\n")
print(summary(aov(MonthlySales ~ Region, data = sales_data)))

# F-Test
cat("\nF-Test:\n")
print(var.test(electronics, accessories))

# Mann-Whitney U-Test
cat("\nMann-Whitney Test:\n")
print(wilcox.test(electronics, accessories))

# Fisher's Exact Test
cat("\nFisher Test:\n")
print(fisher.test(table(sales_data$Category, sales_data$Region)))

# Kruskal-Wallis Test
cat("\nKruskal-Wallis Test:\n")
print(kruskal.test(MonthlySales ~ Region, data = sales_data))

# Bartlett's Test
cat("\nBartlett Test:\n")
print(bartlett.test(MonthlySales ~ Region, data = sales_data))

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# Step 7: Statistical Distributions

# Binomial Distribution
high_sales <- sum(sales_data$MonthlySales > 100)
cat("\nBinomial Probabilities:\n")
print(dbinom(0:5, size = 5, prob = high_sales/length(sales_data$MonthlySales)))

# Poisson Distribution
lambda <- mean(sales_data$MonthlySales) / 10
cat("\nPoisson Probabilities:\n")
print(dpois(0:5, lambda))

# Normal Distribution
cat("\nNormal Distribution (Density):\n")
print(dnorm(sales_data$MonthlySales,
            mean = mean_sales,
            sd = sd_sales))
```
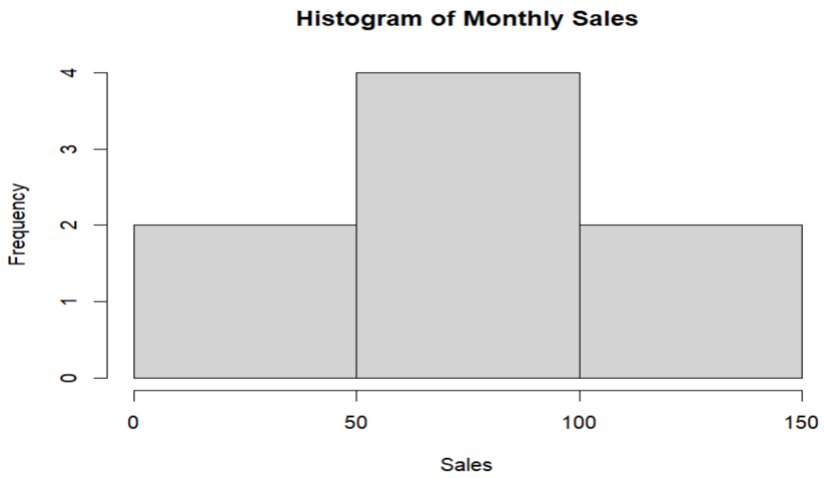
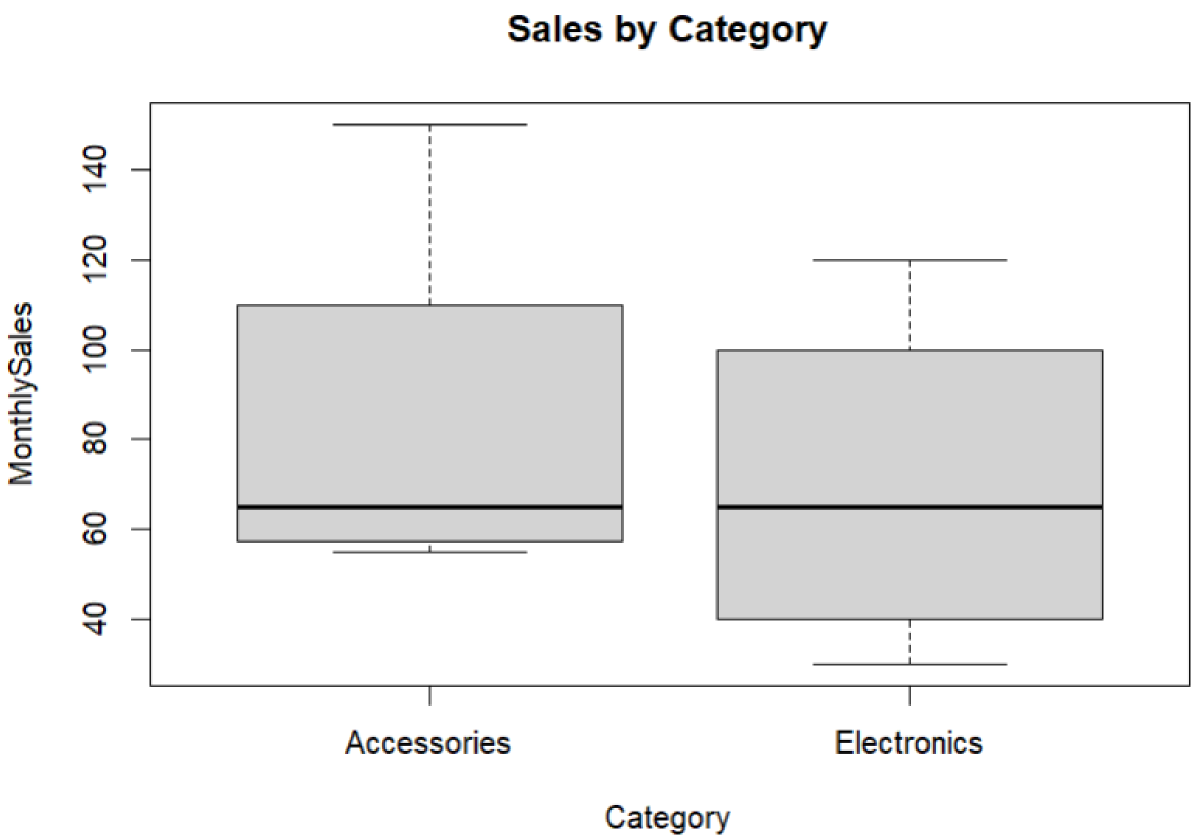

```
Structure:
'data.frame':	8 obs. of  6 variables:
 $ ProductID   : int  1 2 3 4 5 6 7 8
 $ ProductName : chr  "Laptop" "Mobile" "Tablet" "Camera" ...
 $ Category    : chr  "Electronics" "Electronics" "Electronics" "Electronics" ...
 $ Region      : chr  "North" "South" "East" "West" ...
 $ MonthlySales: num  50 120 80 30 150 70 60 55
 $ Price       : num  700 300 250 400 50 20 30 25

Summary:
   ProductID    ProductName          Category            Region           MonthlySales        Price       
 Min.   :1.00   Length:8           Length:8           Length:8           Min.   : 30.00   Min.   : 20.00  
 1st Qu.:2.75   Class :character   Class :character   Class :character   1st Qu.: 53.75   1st Qu.: 28.75  
 Median :4.50   Mode  :character   Mode  :character   Mode  :character   Median : 65.00   Median :150.00  
 Mean   :4.50                                                            Mean   : 76.88   Mean   :221.88  
 3rd Qu.:6.25                                                            3rd Qu.: 90.00   3rd Qu.:325.00  
 Max.   :8.00                                                            Max.   :150.00   Max.   :700.00  

Category Distribution:

Accessories Electronics 
          4           4 

Region Distribution:

 East North South  West 
    2     2     2     2 

Mean: 76.875 
Median: 65 
Mode: 30 50 55 60 70 80 120 150 

Variance: 1563.839 
Standard Deviation: 39.54541 
Range: 30 150 

Correlation (Sales vs Price): -0.3179059 
```

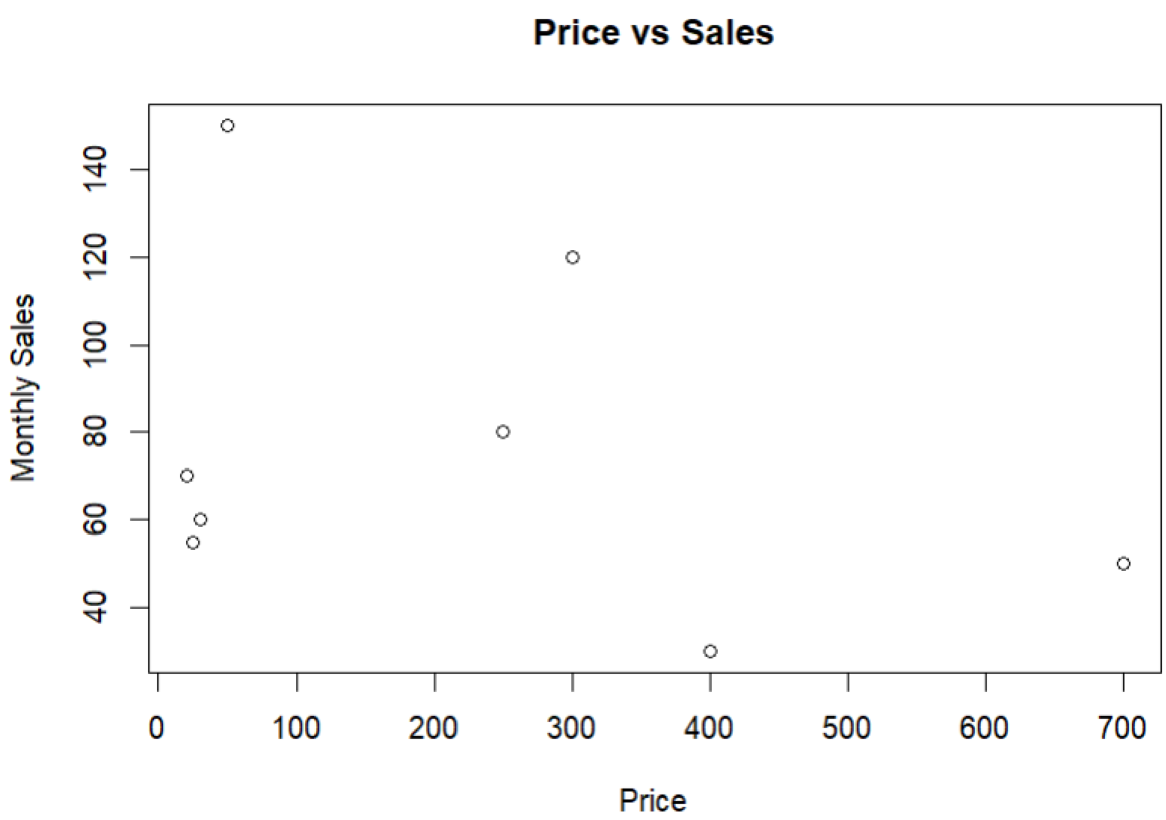
```
Correlation Matrix:
             MonthlySales      Price
MonthlySales    1.0000000 -0.3179059
Price          -0.3179059  1.0000000

T-Test:

	Welch Two Sample t-test

data:  electronics and accessories
t = -0.46332, df = 5.901, p-value = 0.6597
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -86.66347  59.16347
sample estimates:
mean of x mean of y 
    70.00     83.75 


ANOVA:
            Df Sum Sq Mean Sq F value Pr(>F)
Region       3   4184    1395   0.825  0.545
Residuals    4   6763    1691               

F-Test:

	F test to compare two variances

data:  electronics and accessories
F = 0.77068, num df = 3, denom df = 3, p-value = 0.8356
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
  0.04991719 11.89867878
sample estimates:
ratio of variances 
         0.7706806 
```

```
Mann-Whitney Test:

	Wilcoxon rank sum exact test

data:  electronics and accessories
W = 6, p-value = 0.6857
alternative hypothesis: true location shift is not equal to 0


Fisher Test:

	Fisher's Exact Test for Count Data

data:  table(sales_data$Category, sales_data$Region)
p-value = 1
alternative hypothesis: two.sided


Kruskal-Wallis Test:

	Kruskal-Wallis rank sum test

data:  MonthlySales by Region
Kruskal-Wallis chi-squared = 3, df = 3, p-value = 0.3916


Bartlett Test:

	Bartlett test of homogeneity of variances

data:  MonthlySales by Region
Bartlett's K-squared = 2.1462, df = 3, p-value = 0.5426


Binomial Probabilities:
[1] 0.2373046875 0.3955078125 0.2636718750 0.0878906250 0.0146484375 0.0009765625
```

```
Poisson Probabilities:
[1] 0.000458523 0.003524896 0.013548819 0.034718848 0.066725286 0.102590127

Normal Distribution (Density):
[1] 0.008007992 0.005566418 0.010056758 0.004997033 0.001825233 0.009936900 0.009210280 0.008657044
>
```




Sales by Category
MonthlySales
140
120
100
80
60
40
Accessories
Electronics
Category


Price vs Sales
Monthly Sales
140
120
100
80
60
40
0
100
200
300
400
500
600
700
Price